EECS 245, Fall 2026

LEC 16

Projecting onto the Column Space; Regression

→ Read: Ch. 6.3–6.4, 7.1

Agenda

6.3-6.4

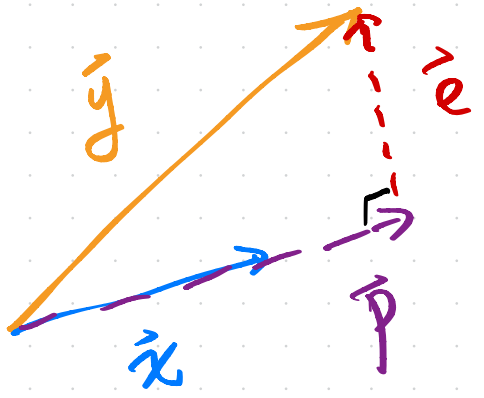
- Projecting onto the column space
  - → Big idea: the normal equation
- Linear regression as a linear algebra problem

7.1

Announcements

- HW 6 due Sunday
- HW 7 out in the next 2 days, due following Sunday
  - → Another supplemental YouTube playlist was linked for today's lecture!

Recap: projecting onto one vector



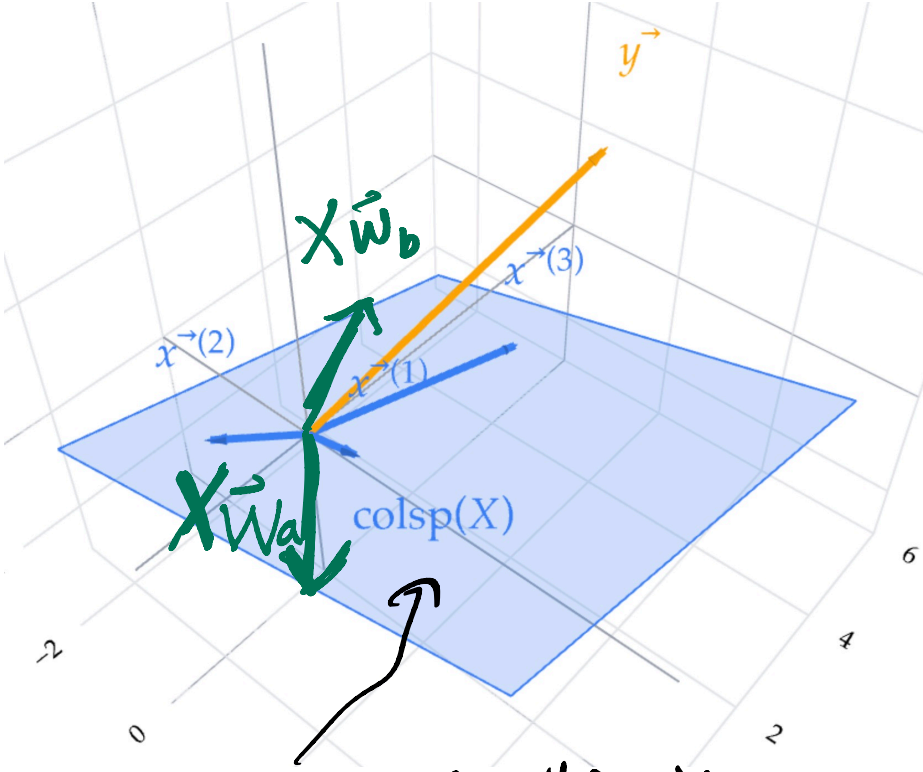
$\vec{p}$: projection of $\vec{y}$ onto $\vec{x}$

$\vec{p} = k^* \vec{x}$, chosen so that

$\vec{e} = \vec{y} - k^* \vec{x}$ orthogonal to $\vec{x}$

Why? the orthogonal error vector is shortest! It minimizes

$$f(k) = \| \vec{y} - k\vec{x} \|^2 \quad \Rightarrow k^* = \frac{\vec{x} \cdot \vec{y}}{\vec{x} \cdot \vec{x}} = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|^2}$$

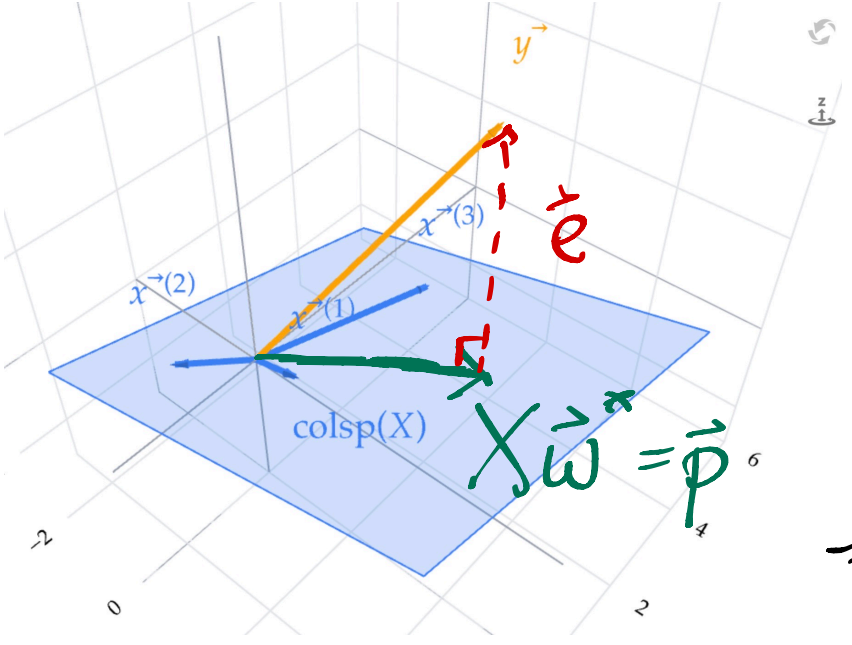


$$\underset{n \times d}{X} = \begin{bmatrix} 1 & 1 & & 1 \\ \vec{x}^{(1)} & \vec{x}^{(2)} & \cdots & \vec{x}^{(d)} \\ 1 & 1 & & 1 \end{bmatrix}$$

$\vec{x}^{(i)} \in \mathbb{R}^n$, $\vec{y} \in \mathbb{R}^n$

Remember, every vector in $\text{colsp}(X)$ is of the form $X\vec{w}$, where $\vec{w} \in \mathbb{R}^d$

in general, this is a $\text{rank}(X)$-dimensional subspace of $\mathbb{R}^n$



Question: what vector $\vec{w}$ makes $X\vec{w}$ as close to $\vec{y}$ as possible?

→ In other words, what $\vec{w}$ minimizes

$$\|\vec{e}\|^2 = \|\vec{y} - X\vec{w}\|^2$$

$X\vec{w}$: lin. comb. of X's columns

$\vec{y} - X\vec{w}$: error vector

Assuming $X, \vec{y}$ fixed: we can only control $\vec{w}$

Goal: choose the $\vec{w}$ that minimizes

$$\|\vec{y} - X\vec{w}\|^2$$

⇒ How? Pick $\vec{w}$ so that

$\vec{y} - X\vec{w}$ is orthogonal to $\text{colsp}(X)$

$\vec{y} - X\vec{w}$: error vector

$\text{colsp}(X)$: a subspace, containing all linear combinations of X's cols

How do we find $\vec{w}$ so that $\underbrace{\vec{y} - X\vec{w}}_{\vec{e}}$ is orthogonal to colsp(X)?

$$\underbrace{X}_{n \times d} = \begin{bmatrix} | & | & & | \\ \vec{x}^{(1)} & \vec{x}^{(2)} & \cdots & \vec{x}^{(d)} \\ | & | & & | \end{bmatrix}$$

$$X \begin{bmatrix} 3 \\ -7 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\vec{e} \cdot \left(3\vec{x}^{(1)} - 7\vec{x}^{(2)}\right)$$

$$= 3\underbrace{\vec{e} \cdot \vec{x}^{(1)}}_{0} - 7\underbrace{\vec{e} \cdot \vec{x}^{(2)}}_{0}$$

$$= 0$$

Idea: if $\vec{e}$ is orthogonal to every column of X individually, it will be orthogonal to colsp(X)!

How do we find $\vec{w}$ so that $\underbrace{\vec{y} - X\vec{w}}_{\vec{e}}$ is orthogonal to $\text{colsp}(X)$?

$$\underbrace{X}_{n \times d} = \begin{bmatrix} | & | & & | \\ \vec{x}^{(1)} & \vec{x}^{(2)} & \cdots & \vec{x}^{(d)} \\ | & | & & | \end{bmatrix}$$

Idea: if $\vec{e}$ is orthogonal to every column of X individually, it will be orthogonal to $\text{colsp}(X)$!

$$\vec{x}^{(1)} \cdot (\vec{y} - X\vec{w}) = 0$$
$$\vec{x}^{(2)} \cdot (\vec{y} - X\vec{w}) = 0$$
$$\vdots$$
$$\vec{x}^{(d)} \cdot (\vec{y} - X\vec{w}) = 0$$

} system of d equations, d unknowns

$$\vec{x}^{(1)} \cdot (\vec{y} - X\vec{w}) = 0$$
$$\vec{x}^{(2)} \cdot (\vec{y} - X\vec{w}) = 0$$
$$\vdots$$
$$\vec{x}^{(d)} \cdot (\vec{y} - X\vec{w}) = 0$$

} system of d equations, d unknowns

How can we represent as a single equation?

$$X^T(\vec{y} - X\vec{w}) = \vec{0}$$

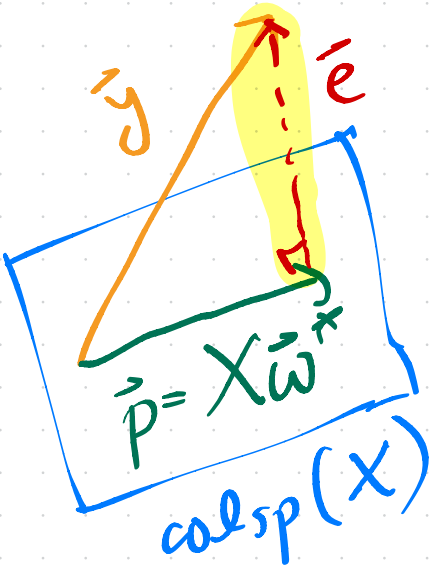
rows of X^T are cols of X !!!

Aside: two interpretations

$A\vec{v}$

(1) dot products of A's rows with $\vec{v}$

(2) linear combination of A's cols, with coefficients from $\vec{v}$



Goal: find $\vec{w}$ such that $\vec{y} - X\vec{w}$ orthogonal to colsp(X)

Equivalent to finding $\vec{w}$ such that

$$X^T(\vec{y} - X\vec{w}) = \underbrace{\vec{0}}_{\in \mathbb{R}^d}$$

$$X^T\vec{y} - X^TX\vec{w} = \vec{0}$$

$$X^TX\vec{w} = X^T\vec{y}$$

"normal equation(s)"

because $\vec{e}$ is perpendicular/normal

All that's left is finding $\vec{w}$ such that

$$X^T X \vec{w} = X^T \vec{y}$$

square matrix!

Case 1: $X^T X$ is invertible : happens when X's cols are linearly independent!

$$\underbrace{(X^T X)^{-1} X^T X}_{I} \vec{w} = (X^T X)^{-1} X^T \vec{y}$$

$$\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$$

most important equation of the semester

When is X^TX invertible?

X : $n \times d$ $\quad$ X^TX : $d \times d$

For X^TX to be invertible, we need $\text{rank}(X^TX) = d$

=> Important: $\text{rank}(X^TX) = \text{rank}(X)$! why? see video on website

We need X^TX invertible => we need $\text{rank}(X^TX) = d$

=> $\text{rank}(X) = d$ => we need X's columns to be linearly independent!

Case ②: X^TX not invertible

(i.e. X's cols are <u>not</u> linearly independent)

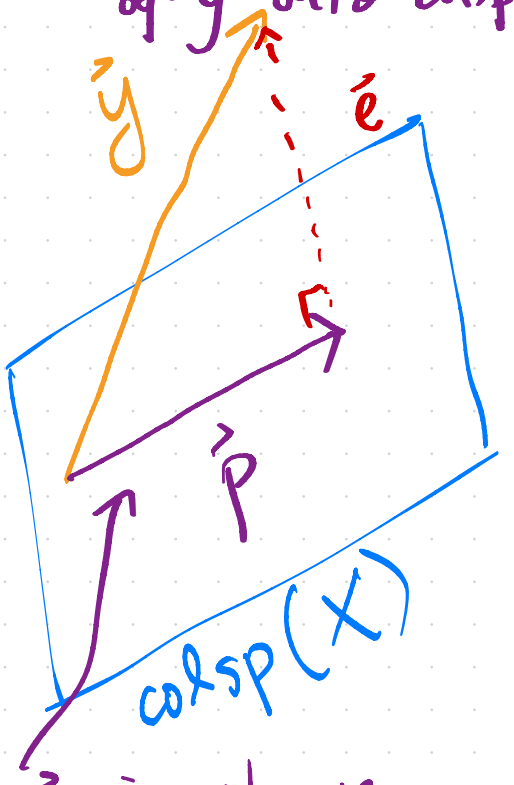
then, $$X^TX\vec{w} = X^T\vec{y}$$

has infinitely many solutions $\vec{w}^*$!

→ HW 7 will show you how

→ involves finding nullsp(X) and deleting redundant cols

$\vec{p}$: orthogonal projection of $\vec{y}$ onto colsp(X)



$\vec{p}$ is always unique!

No matter what, $\vec{p}$ is unique!

⇒ If X's cols are lin. ind., then only one lin. comb. of X's cols $= \vec{p}$

$$\vec{p} = X\vec{w}^*$$

($\vec{w}^*$: unique)

⇒ Otherwise, many different $\vec{w}^*$'s give the same $\vec{p}$!

Example

$$X = \begin{bmatrix} 1 & 0 \\ 2 & 1 \\ 1 & 1 \\ 0 & -1 \end{bmatrix} \qquad \vec{y} = \begin{bmatrix} 1 \\ 0 \\ 4 \\ 5 \end{bmatrix}$$

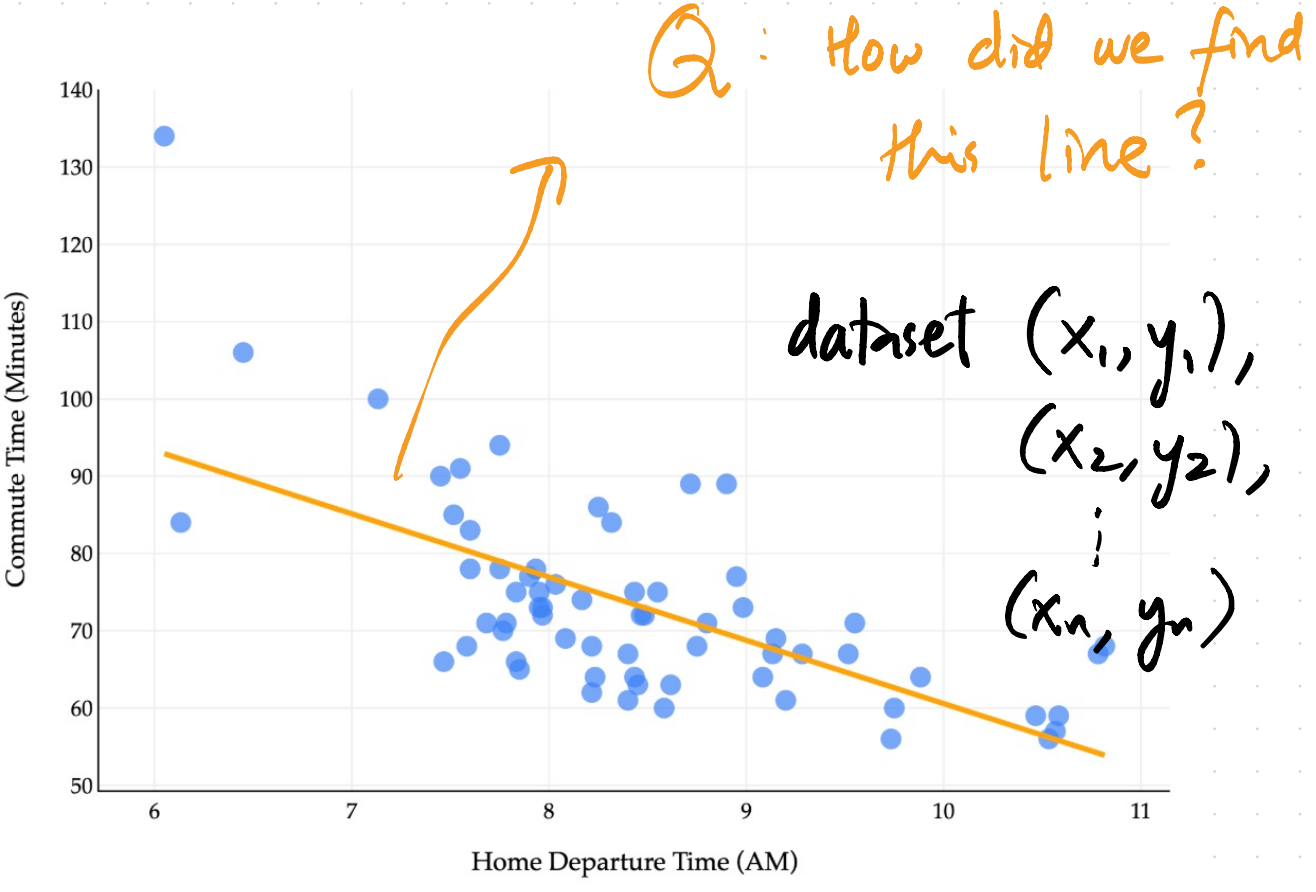
Q: which linear combination of X's cols is closest to $\vec{y}$?

A: $X\vec{w}^*$, where $\vec{w}^* = (X^TX)^{-1}X^T\vec{y} = \begin{bmatrix} 2 \\ -8/3 \end{bmatrix}$

among all of the linear combinations of X's cols, the vector closest to $\vec{y} = \begin{bmatrix} 0 \\ 4 \\ 5 \end{bmatrix}$ is

$$\hat{p} = X\vec{w}^* = (2)\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + (-8/3)\begin{bmatrix} 0 \\ 1 \\ 1 \\ -1 \end{bmatrix}$$

$\vec{x}^{(1)}$ $\vec{x}^{(2)}$


Q: How did we find this line?
dataset (x_1, y_1),
(x_2, y_2),
⋮
(x_n, y_n)
Commute Time (Minutes)
140
130
120
110
100
90
80
70
60
50
6
7
8
9
10
11
Home Departure Time (AM)

Three-step modeling recipe

1. Choose a model: $h(x_i) = w_0 + w_1 x_i$ (intercept: w_0; slope: w_1)
2. Choose a loss function: squared loss
3. Minimize average loss (empirical risk) to find optimal parameters

"mean squared error"

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \underbrace{(w_0 + w_1 x_i)}_{h(x_i)} \right)^2$$

n: # data points

$$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \underbrace{(w_0 + w_1 x_i)}_{h(x_i)} \right)^2$$

→ to find w_0^*, w_1^*, we took partial derivatives and solved

→ annoying if we had more features and parameters

→ goal: rewrite R_{sq} using linear algebra!

$$\vec{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

"observation vector"

e.g. actual commute times

"design matrix" X

$$\vec{p} = \begin{bmatrix} h(x_1) \\ h(x_2) \\ \vdots \\ h(x_n) \end{bmatrix} = \begin{bmatrix} w_0 + w_1 x_1 \\ w_0 + w_1 x_2 \\ \vdots \\ w_0 + w_1 x_n \end{bmatrix} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}_{n \times 2} \begin{bmatrix} w_0 \\ w_1 \end{bmatrix}$$

"prediction vector"

$\vec{p} \in \mathbb{R}^n$

$\vec{w}$

"parameter vector"

before: $$R_{sq}(w_0, w_1) = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - (w_0 + w_1 x_i)\right)^2$$

now: $$R_{sq}(\vec{w}) = \frac{1}{n} \| \vec{y} - X\vec{w} \|^2$$

error vector, $\vec{e}$!

For the $n \times 2$ design matrix, $(X^T X)^{-1}$ exists, so $\vec{w}^* = (X^T X)^{-1} X^T \vec{y}$